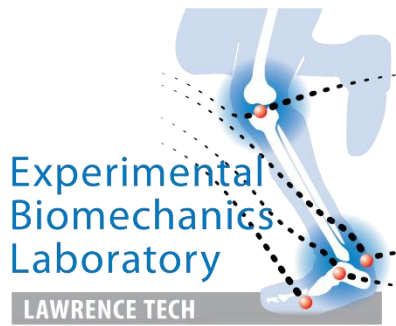
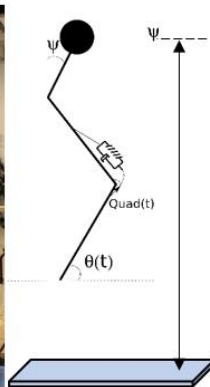
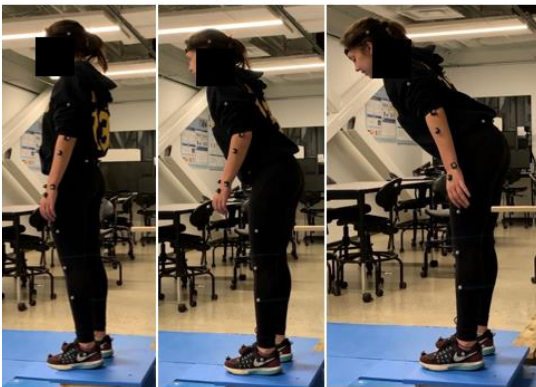




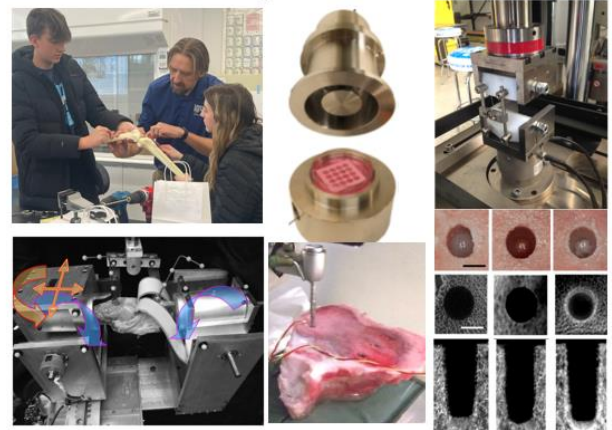
The [Experimental Biomechanics Laboratory \(EBL\)](#) at LTU is located in J134, Taubman Complex. This teaching and research lab has open/flexible space and state of the art motion tracking system and other biomechanics equipment for studying many human movement activities. The Vicon 3D optical motion analysis system includes 10 Bonita cameras, and a Basler high speed reference video camera, with Nexus, Polygon and Plug-In-Gait software. Two Kistler force platforms are digitized by the Vicon system to determine the relevant Ground Reaction Force (GRF) vectors and center of pressure on the feet. A 16-channel Delsys Trigno, wireless sensor system with a variety of physiological monitoring sensors, including; surface electromyography (EMG),

3-axis accelerometers and electrocardiography (EKG) electrodes. Data is transmitted to the base station using a 2.4 GHz RF communication protocol and then recorded and analyzed with Delsys EMGworks software, or synced through the Vicon system to simultaneously record the 3D motions, GRF and EMG simultaneously during a human biomechanics experiment. The Vicon system and forceplates are portable or can be mounted in place to create a variety of capture volume configurations. Alternatively, a system of 15 mBient (model MMRL) wearable inertial measurement unit (IMU) sensors can be used to take measurements inside or outside of the laboratory. Each sensor consists of; 3D accelerometer, gyroscope, magnetometers, a battery, memory storage and can either stream data in real time via Bluetooth wireless connection or log data for later downloading. Sensor fusion is used to determine each of the body segment orientations. Recently, markerless motion tracking using machine learning based computer vision systems allow human movement tracking and facial analysis from standard video systems. Two eye tracker systems are also available, one for stationary recording in conjunction with a computer display as well as a portable unit for real world research collection. For any motion capture system, the joint kinematics and kinetics are analyzed using inverse dynamics methods with a custom human biomechanical model.

Injury Mechanisms



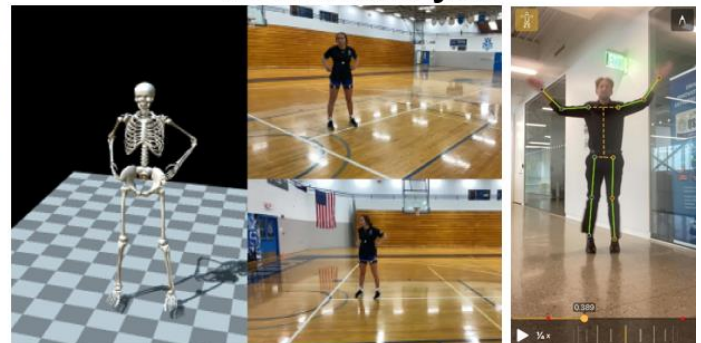
Orthopedics



Wearable Technology



Motion Analysis



with the goal to advance experimental biomechanics understanding by providing practical training to engineering and premedical students and to advance the boundary of biomechanics knowledge through translational research and wearable technology design. To achieve these goals, he combines in vivo testing of human and animal subjects with cadaver and surrogate in vitro models and computer simulations to understand normal health, acute injury and degenerative mechanisms. Meyer enthusiastically cultivates collaborations with clinicians and the medical device industry to develop preventative and regenerative treatments for bone and soft tissue damage and disease.

Recently, he co-founded a multidisciplinary [Wearable Technology Innovation Center \(WTIC\)](#) that provides practical, hands-on experiences to students focused around the topics of sensing, perception, and control in next generation robotics, haptics, quantified-self, and protective devices. The WTIC is a partnership between more than 20 LTU faculty and staff and more than 30 regional industry, academic, non-profit and government partners. The WTIC promotes Open Innovation for Human Enhancement and works with its partners and members of the community to coalesce and support a thriving wearable technology innovation ecosystem in Southeast Michigan. His research focus on preventing sports injuries also overlaps with engineering design projects that allow youth and recreational athletes to track their biomechanics during movement to customize their training using wearable sensors.

Dr. Meyer has published one book chapter and 42 peer-reviewed journal and conference proceeding articles that have received more than 2250 peer citations. He has given more than 100 conference, workshop and invited presentations, including as the [Presidential Colloquium](#) speaker for the [2023 Lawrence Technological University Research Day](#) and as a panel member for recent Humanities + Technology topics such as “AI and the Future of Education”. Working with Dr. Salah Huwais, a practicing Periodontist, inventor and founder of Versah LLC they presented and published a seminal biomechanical validation study of [Osseodensification](#), in the International Journal of Oral and Maxillofacial Implants a novel implant osteotomy preparation technique that increases implant stability. Dr Meyer is an invited speaker for the session “Biomechanical Evidence” at the 2017 *Osseodensification World Symposium*, in Orlando, FL.

As an Associate Professor of Biomedical Engineering, Dr. Meyer teaches undergraduate required courses; BME Best Practices, Introduction to Biomechanics and Biomechanics Lab, and Tissue Mechanics, as well as the elective Engineering Applications in Orthopedics and the graduate course Cellular and Molecular Mechanobiology. He has also been an active member of the College of Engineering faculty committee that is redesigning required courses for all students as a part of a cohesive, four-year multidisciplinary engineering design program focused on creating entrepreneurially minded engineers. Dr. Meyer has developed entrepreneurial-minded learning course modules and trained other faculty to use similar modules in their courses as part of an ongoing Kern Family Foundation grant that he received in 2013. Educational materials developed by this network of faculty are shared at the website; [engineeringunleashed.com](#). In addition, Dr. Meyer offers a number of outreach programs for high school students and advises many projects for undergraduate and graduate students.

Meyer received his PhD in Mechanical Engineering (Biomechanics) from Michigan State University. The primary topic of his research was lower extremity biomechanics during sports injury. He then completed his post-doctoral training at the Centre for Bioengineering at Trinity College Dublin, Ireland in cartilage tissue engineering and mechanobiology. He was an invited speaker in the session “Solving the mystery of non-contact ACL injuries - how do they happen?” at the 2011 *IOC World Conference on Prevention of Injury & Illness in Sport*, in Monte-Carlo, Monaco.

Experimental Biomechanics Laboratory (EBL)

The EBL is a 1300 sq. ft. facility located in J134, Taubman Complex. This teaching and research lab has open/flexible space and state of the art motion tracking system and other biomechanics equipment for studying many human movement activities. The Vicon 3D optical motion analysis system includes 10 Bonita cameras, and a Basler high speed reference video camera, with Nexus, Polygon and Plug-In-Gait software. Two Kistler force platforms are digitized by the Vicon system to determine the relevant Ground Reaction Force (GRF) vectors and center of pressure on the feet. A 16-channel Delsys Trigno, wireless sensor system with a variety of physiological monitoring sensors, including; surface electromyography (EMG), 3-axis accelerometers and electrocardiography (EKG) electrodes. Data is transmitted to the base station using a 2.4 GHz RF communication protocol and then recorded and analyzed with Delsys EMGworks software, or synced through the Vicon system to simultaneously record the 3D motions, GRF and EMG simultaneously during a human biomechanics experiment. The Vicon system and forceplates are portable or can be mounted in place to create a variety of capture volume configurations.

Wearable Technology Innovation Center (WTIC)

The WTIC has maker-space facilities in J353/J355, Taubman Complex. The Center is a multidisciplinary collaboration between various engineering disciplines, industrial design, natural science, computer science, social science and other academics at LTU. The partners provide practical, hands-on experiences to students

focused around the topics of sensing, perception, and control in next generation robotics, haptics, and other quantified-self devices. Design projects are focused on developing consumer grade wearable sensors and portable systems for research and teaching applications. Entrepreneurship and innovation partnerships are encouraged to build commercial opportunities with the Centropolis Accelerator and other local industry and start-up companies. The Center shares equipment with the Biomedical Engineering Department, including; desktop computers, virtual reality systems, 3D printers, a small CNC milling machine, laser cutter, vinyl cutter, sewing machines, soldering and surface mount electronics fabrication, power supplies, oscilloscopes, multimeters, data acquisition equipment and various microprocessors, sensors and actuators.

Funding for the EBL equipment, supplies and personnel come from a variety of sources including; internal support from LTU, foundation grants, in kind donations from industry, and contract research project with industry partners. The EBL has supported LTU's efforts for increased partnership with local hospitals such as Beaumont, Detroit Medical Center, Henry Ford and Providence. These projects are currently being developed in the areas of oral and orthopaedic surgery, sports medicine, simulations, radiology and robotic surgery. The EBL has trained/supervised more than 22 undergraduate student research projects, 13 BME senior projects, and 5 graduate student research projects.